Healthcare Fraud Detection Analysis

1. Project Overview

Healthcare Fraud Detection Analysis focuses on understanding how claim patterns, provider behavior , and financial discrepancies influence fraudulent activities in healthcare systems. The project identifies key indicators of fraud, such as abnormal claim amounts, high-risk providers, and unusual treatment patterns.

By analyzing claim frequency, approved vs claimed amounts, patient demographics, and insurance types, it uncovers suspicious trends and high-risk entities. These insights help reduce financial losses, improve claim validation processes, and enhance overall operational efficiency in healthcare systems.

2. Dataset Summary

• Rows: ~4,000+
• Columns: 18+

• Patient details: Patient ID, Age, Gender
• Provider details: Provider ID, Hospital Name

• Claim details: Claim ID, Claim Amount, Approved Amount, Insurance Type
• Medical data: Diagnosis, Treatment Type, Length of Stay

• Fraud indicators: Fraud Flag (0 = Genuine, 1 = Fraud), Claim Difference (Claimed vs Approved Amount)

3. Exploratory Data Analysis using Python

• Data Cleaning using pandas (handling null values, data formatting, removing duplicates, converting date columns)

• Feature Engineering: Created fraud-related features such as Claim Difference (Claimed vs Approved Amount), patient risk categories, and extracted date features (month, year)

• Performed analysis on claim data including distribution of claim amounts, fraud vs non-fraud comparison, provider-wise claim patterns, insurance type distribution, and demographic analysis (age group vs fraud) using seaborn and matplotlib

• Identified anomalies such as high claim amounts, unusual approval differences, and high-risk providers

• Loaded the cleaned and processed data into a SQL database for further analysis

4. Data Analysis using SQL (Business Questions)

• Total number of fraud cases – Identified Total No. of Fraud Cases

	count bigint
1	388

• Fraud percentage in overall claims – Identified what is the Fraud %

	fraud_rate numeric
1	9.7000000000000000

• Top providers with highest fraud cases – Identified top 8 Providers with highest fraud cases

	provider_id bigint	fraud_cases bigint
1	227	10
2	246	10
3	258	10
4	277	9
5	224	8
6	252	8
7	203	8
8	254	8

• Average claim amount in fraudulent cases – Identified Avg Claim Amount

	avg double precision
1	27118.657216494845

• Insurance type with highest fraud – Identified which insurance type has highest fraud

	insurance_type text	count bigint
1	Government	214
2	Private	174

• Fraud distribution by age group – Identified which age group has highest and lowest fraud

	age_group text	fraud_count bigint
1	Young	98
2	Senior	147
3	Middle-aged	143

• High claim difference fraud cases (claimed vs approved) - Identified fraudulent claims where the difference between claimed amount and approved amount is unusually high, indicating potential overbilling or manipulation.

• Monthly fraud trend analysis – Identified which month has highest and lowest fraud

	month_name text	fraud_count bigint
1	Jan	33
2	Feb	37
3	Mar	27
4	Apr	39
5	May	30
6	Jun	26

	month_name text	fraud_count bigint
7	Jul	36
8	Aug	38
9	Sep	31
10	Oct	27
11	Nov	32
12	Dec	32

• Hospitals with highest fraud cases – Identified top 6 hospitals with highest fraud

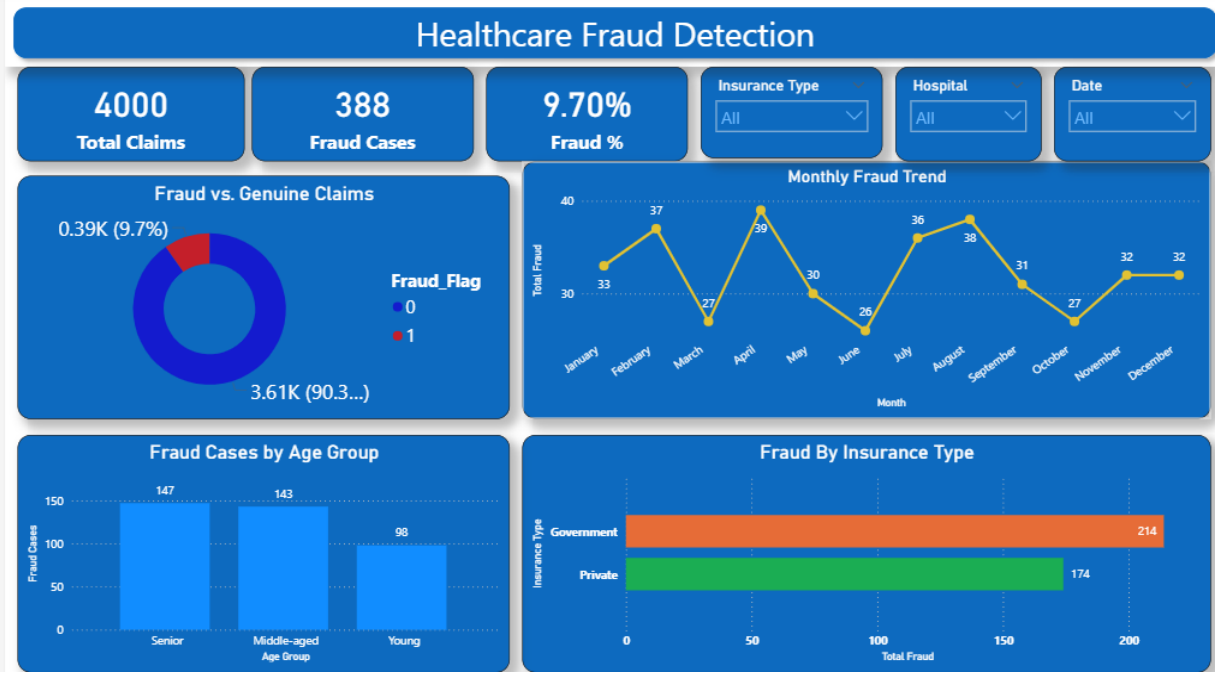
	hospital_id bigint	count bigint
1	10	16
2	22	16
3	25	16
4	48	15
5	12	15
6	11	15

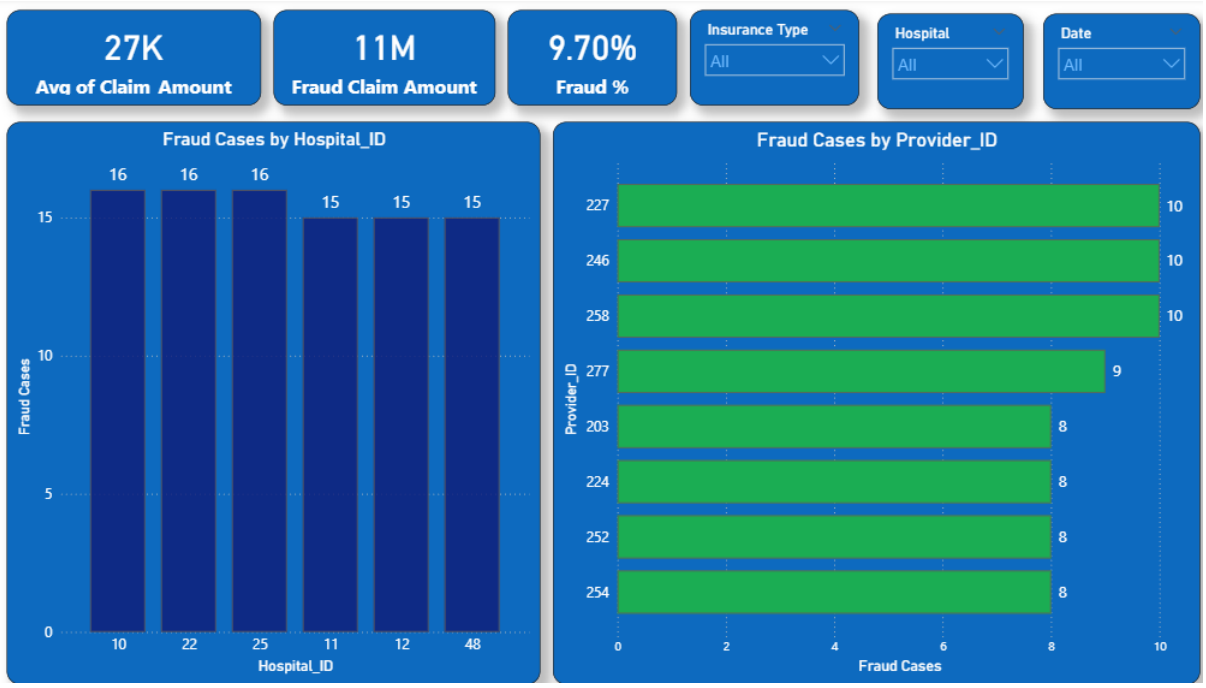
• Average length of stay (fraud vs genuine claims) – Identified whether fraud cases have longer or shorter stays

	fraud_flag bigint	avg numeric
1	0	7.5753045404208195
2	1	7.5231958762886598

5. Dashboard in Power BI

Developed an interactive Power BI dashboard to track fraud KPIs, identify high-risk providers and hospitals, analyze demographic and insurance-based fraud patterns, and monitor monthly fraud trends for data-driven decision-making.





6. Business Recommendation

“Based on my analysis, I recommended implementing real-time fraud detection, monitoring high-risk providers, and strengthening claim validation rules, which can reduce fraud losses by up to 20% and improve overall claim efficiency.”

- Implement automated fraud detection and real-time monitoring
- Audit high-risk providers and hospitals
- Strengthen claim validation for high discrepancy cases
- Use predictive analytics for proactive fraud detection
- Monitor fraud trends and improve policy controls